

UCE512: DESIGN OF STEEL STRUCTURES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To learn (i) basic concepts and applications of various types of soil reinforcement, and (ii) design of reinforced earth structures				
Introduction: Loads, structural steels and their specifications, structural elements, design specifications as per IS: 800, structural layout, strength and stiffness considerations, efficiency of cross-section, safety and serviceability considerations as per IS2062-2011 Bolted and Welded Connection: Types of Bolts, advantages and disadvantages of bolted connections, failure of bolted joints, design of the concentric bolted joint, the efficiency of joint. Design of Eccentric bolted connections. Welded Connection: Types of welded joints, design of welded joint subjected to axial loads Tension Members: Types of tension members, net area, net effective area for angles, tees, design of tension members, tension splice, and lug angles. Compression Members: Axially loaded columns, effective length, slenderness ratio, allowable stresses, general specifications, design of axially loaded members, laced and battened columns and their design, built-up compression members Plastic Design: Introduction, advantages and disadvantages, , theory of plastic bending, plastic hinge mechanism, collapse load analysis, static and mechanism method, distributed loading, design consideration. Flexural Members (Beams): Design criteria, permissible stresses, laterally supported beams and their design laterally unsupported beams and their design, web buckling, web crippling				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Design tension members 2. Design the bolted and welded connections between various structural components 3. Design compression members 4. Find out ultimate load of the structural systems using plastic analysis 5. Design flexural members				
Text Books 1. Subramanya, N, Design of Steel Structures, N. Subramanian, Oxford University Press (2016). 2. Duggal, S.K. Limit State Design of Steel Structures, McGraw Hill (2010.) 3. Bhavikatti S.S, Design of Steel Structures, Ik International Publishing House, New Delhi,				

2017

4. Sai Ram, K.S, Design of Steel Structures, Pearson Education in South Asia, 2010.
5. Chandra R. and Gehlot V, Limit State Design of Steel Structures, Scientific Publishers, 2009

Reference Books

1. Ajmani, A. L. and Arya, A. S., Design of Steel Structures, Nem Chand and Brothers (2000).
2. Dunham, C.W., Planning of Industrial Structures, John Wiley and Sons (2001).
3. Gary, W., Steel Designer's Manual, Prentice Hall (2008).
4. Gambhir M L, Fundamentals of Structural Steel Design, McGraw Hill Education India Pvt Limited, 2013.
5. Sarwar Alam Raz, Structural Design in Steel, New Age International Publishers, 2014

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25